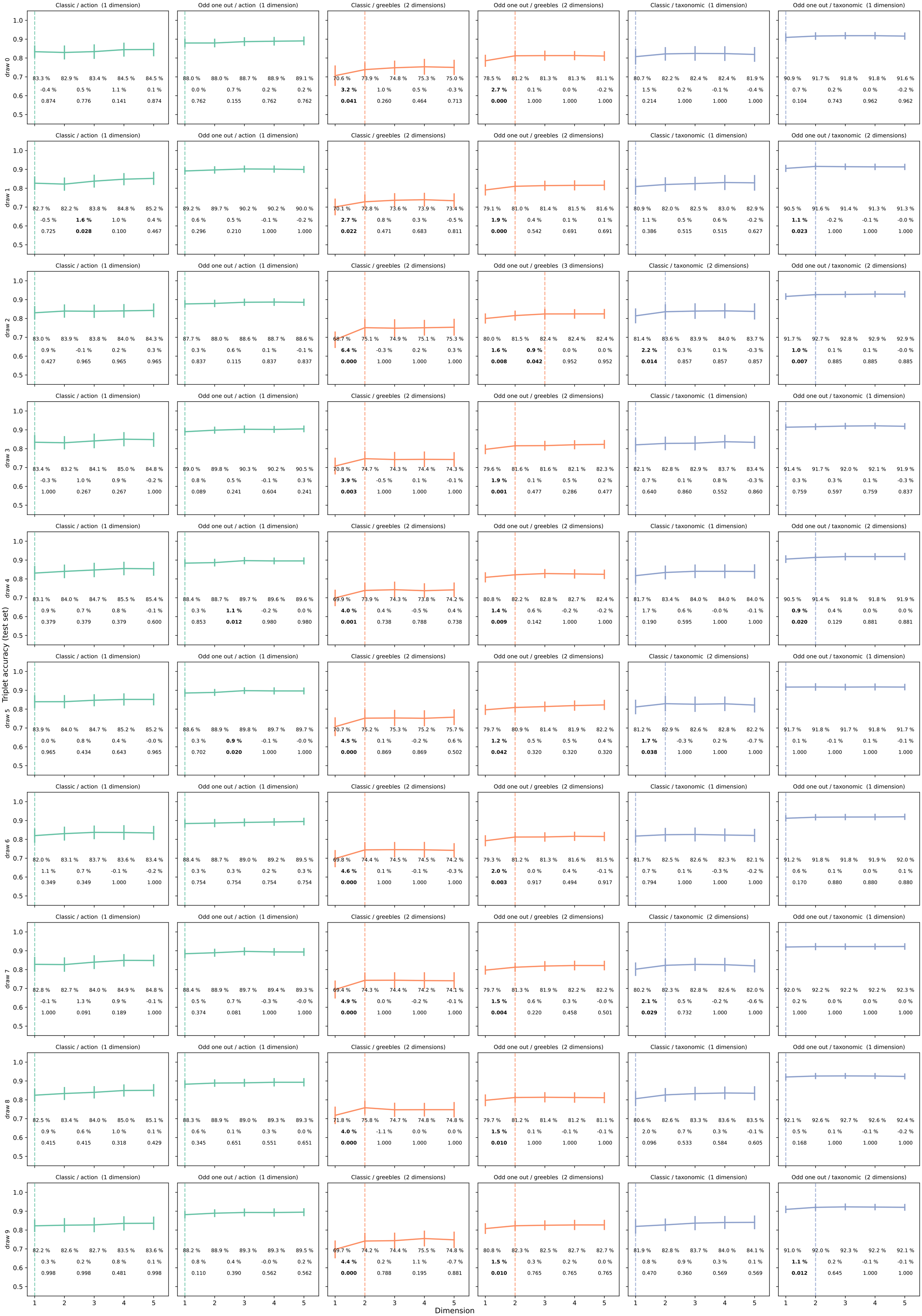design A: scaled-down pool